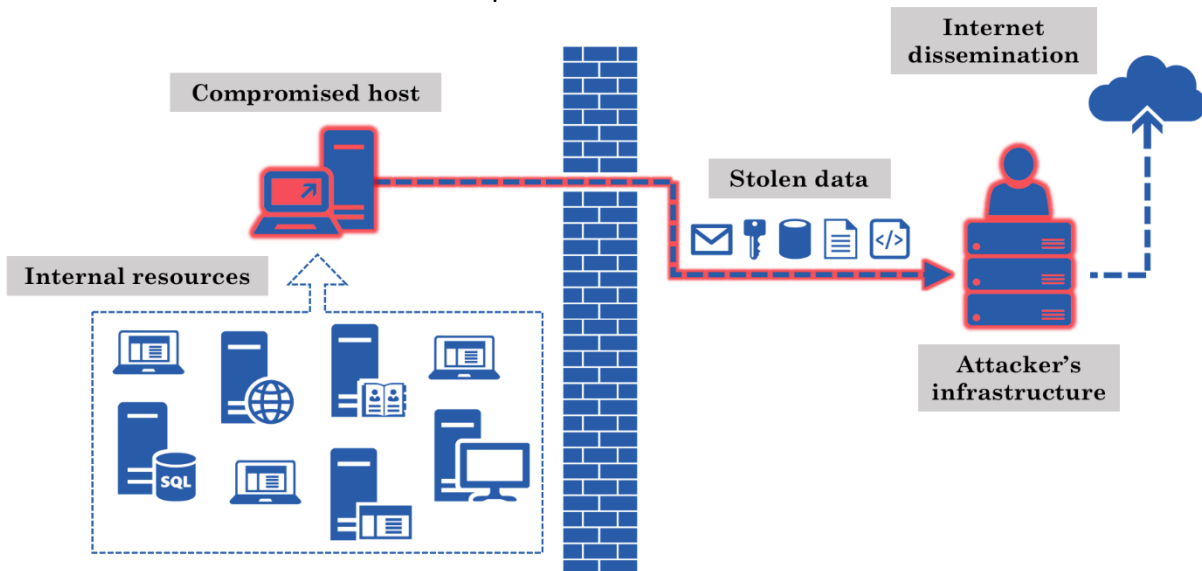


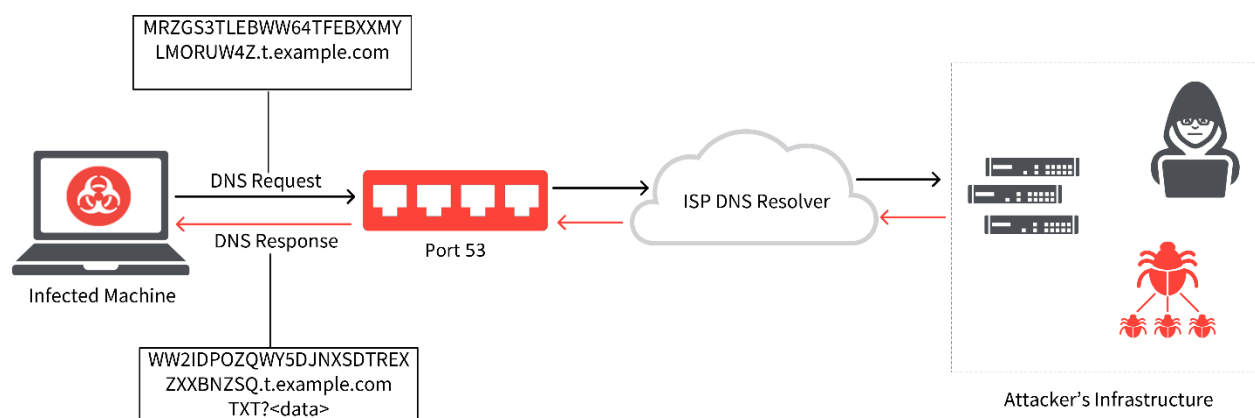
Data Exfiltration:

- o Data exfiltration is unauthorized copying, transfer or retrieval of data from a computer.
- o Data exfiltration is the malicious activity performed through various different techniques.
- o Data exfiltration activity is typically, by cybercriminals over the Internet or other network.
- o Data exfiltration is also known as the data extrusion, the data exportation or the data theft.
- o Data exfiltration, data loss or data leakage is essentially the theft or removal of information.
- o Data from electronic device, typically computer, mobile phone or other Internet connected.
- o This typically includes valuable personal or corporate information that can be sold or used.
- o It is big business for cybercriminal & nightmare for organization that find themselves victim.
- o The Data Exfiltration is very difficult to detect because it happens silently in the background.
- o Victim does not realize it has occurred, leave organizations & individuals highly vulnerable.
- o People are surprised how much data their mobile device consumes overnight even not use.
- o Most of the time this is commonly data being exfiltrated from his/her device or computer.
- o It occurs when malware and/or a malicious actor carries out an unauthorized data transfer.
- o The Data exfiltration or the data exportation is also considered to be a form of data theft.



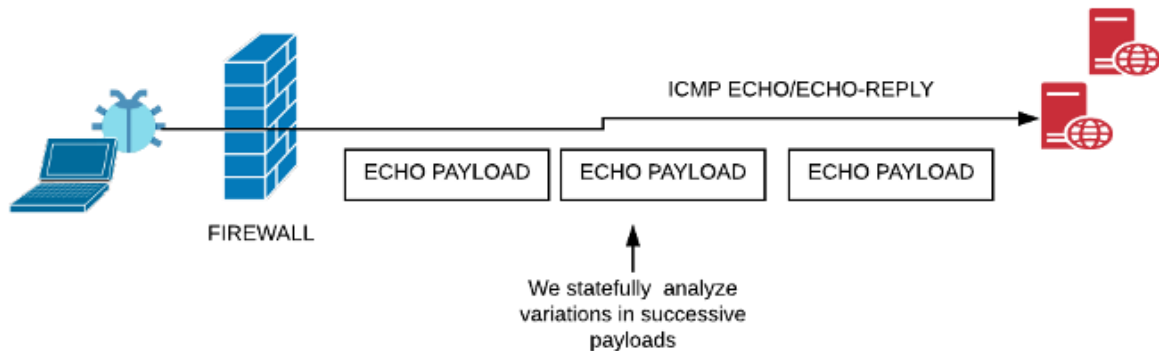
DNS Tunneling:

- o The DNS protocol is main & core component of the Internet protocol (IP) suite.
 - o Domain Name System main goal is the translation of hostnames to IP addresses.
 - o DNS Tunneling is a method of cyber-attack that encodes data of other programs.
 - o That encodes data of other program or protocols in DNS queries and responses.
 - o DNS tunneling includes data payloads that can be added to an attacked DNS server.
 - o DNS tunneling includes data payload used to control remote server & applications.
 - o DNS tunneling requires compromised system to have external network connectivity.
 - o The DNS tunneling requires access to an internal DNS server with network access.
 - o Cybercriminals know that DNS is the well-established, well-known & trusted protocol.
 - o They Know that many organizations do not examine DNS traffic for malicious activity.
 - o DNS tunneling enables these cybercriminals to insert malware or pass stolen information.
 - o Pass DNS queries, creating a covert communication channel that bypasses most firewalls.
 - o It exploits DNS protocol to tunnel malware and other data through a client-server model.
 - o DNS tunneling is technique which exploits DNS protocol for tunneling data via DNS query.
1. The attacker registers a domain, such as badsite.com. The domain's name server points to the attacker's server, where a tunneling malware program is installed.
 2. The attacker infects a computer, which often sits behind a company's firewall, with malware. Because DNS requests are always allowed to move in and out of the firewall, the infected computer is allowed to send a query to the DNS resolver. The DNS resolver is a server that relays requests for IP addresses to root and top-level domain servers.
 3. The DNS resolver routes the query to the attacker's command-and-control server, where the tunneling program is installed. A connection is now established between the victim and the attacker through the DNS resolver. This tunnel can be used to exfiltrate data or for other malicious purposes. Because there is no direct connection between the attacker and victim, it is more difficult to trace the attacker's computer.



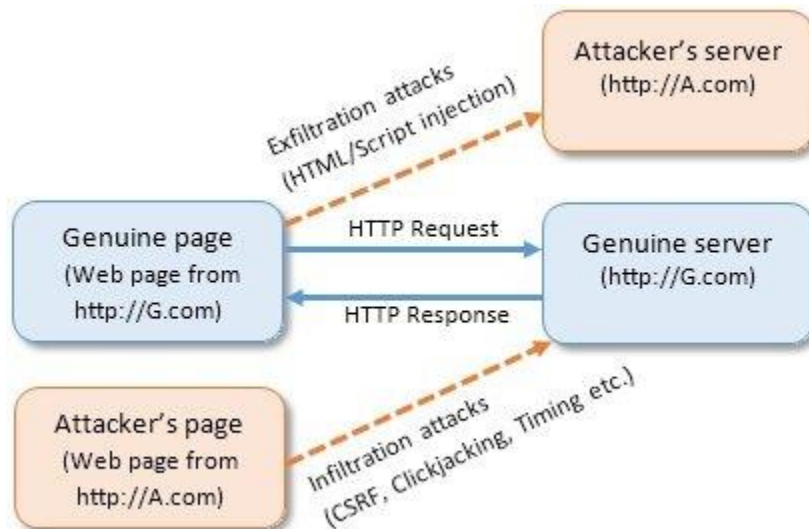
ICMP Tunneling:

- o ICMP stand for Internet Control Message Protocol (ICMP) is an error-reporting protocol.
- o Internet Control Message Protocol used in network devices such as routers and servers.
- o ICMP is fundamental protocol used throughout Internet & part of the Internet protocol.
- o Every IP network device has the capability to send, receive, and process ICMP messages.
- o ICMP tunneling works by injecting arbitrary data into an echo packet sent to remote PC.
- o Remote PC replies in the same manner, injecting an answer into another ICMP packet.



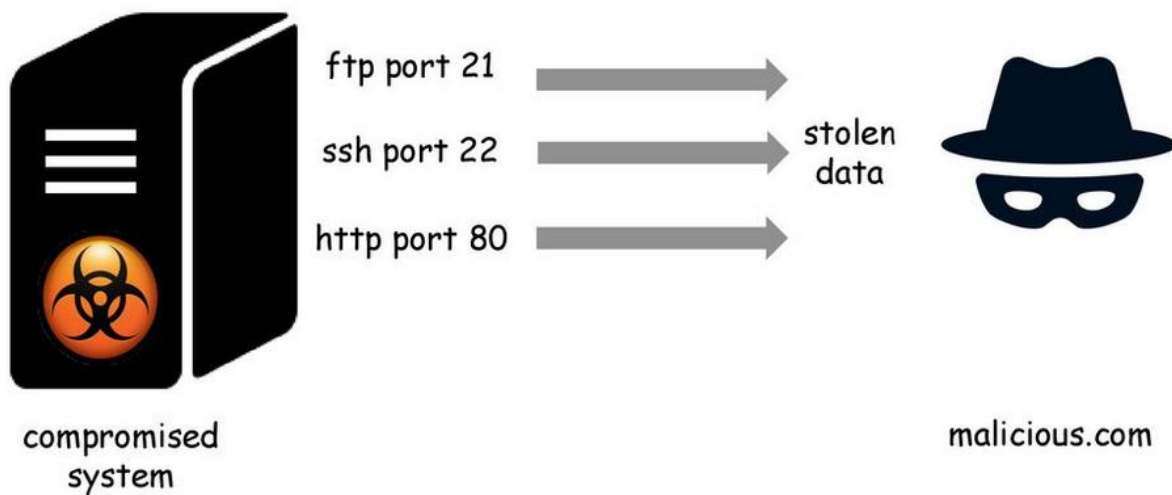
HTTPS:

- o Hypertext Transfer Protocol Secure are the most commonly used exfiltration channels.
- o Traffic exfiltrated over these connections is difficult to distinguish from legitimate traffic.
- o Cybercriminals use Hypertext Transfer Protocol HTTP as a source for data exfiltration.
- o It uses ports like 80 and 443 (for HTTP or HTTPS) and port 53 (for DNS) to hide traffic.



FTP/SSH/SCP/SFTP:

- o FTP/SSH/SCP/SFTP might be permitted outbound and used as data exchange protocols.
- o Over the network medium commonly used protocols such as FTP/SSH/SCP/SFTP etc.
- o To perform data exfiltration using FTP, adversaries authenticate to external FTP server.
- o Inefficient firewall rules in organizations' networks are unable to prevent outbound.
- o Allow hackers to easily establish a connection back to their own malignant infrastructure.
- o Additionally, most operating systems like Windows OS include the default FTP client.
- o Therefore, malicious actors don't have to install any additional tools on exploited host.
- o Hackers enter into externally accessible FTP server of organization's compromised host.
- o Hackers can also attempt to configure the FTP server with write-only permissions.



Email:

- o Outbound email can be used to exfiltrate email, databases, planning documents.
- o Outbound email can be used to exfiltrate email calendars, images and any object.
- o Data can be transmitted to third party as an email or text message or file attachment.
- o Forwarding rule in Microsoft Outlook that enables hackers to receive copies of emails.